

NICKOLAS SIMONS

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science

Minor in Game Studies and Design

Urbana, IL

Aug. 2022 – May 2025

GPA: 3.96

EXPERIENCE

The stu/dio at Illinois

Programming Lead

Urbana, IL

Aug. 2024 – Dec. 2025

- Managed a team of 12 programmers across 8 projects, conducting code reviews and providing planning feedback to ensure code quality and maintainability
- Maintained Perforce infrastructure and integrated Horde for automated builds and Helix Swarm for code review
- Established coding standards and development workflows to improve code maintainability across projects

Gameplay Programmer

April 2024 – Dec. 2025

- Designed and implemented gameplay features and tools for VR platforms using C++ and Blueprints, with an emphasis on accessibility, performance, and usability in classroom environments
- Debugged and resolved critical gameplay issues affecting interaction flow, core mechanics, and level progression
- Optimized gameplay code and materials to meet VR performance targets using Unreal Insights and Nvidia Nsight

PROJECTS

Master Dancer VR | UE5, C++, Blueprint

May 2024 – Dec 2025

- Built three quantized movement-based rhythm games end-to-end, developing core gameplay architecture using C++ and Blueprint
- Created graphical level authoring tools to streamline designer workflows and support rapid iteration
- Implemented a system for anchoring and rendering subtitles in 3D space for VR applications, with support for localization
- Developed modular gameplay systems with a focus on performance, reusability, and extensibility

VRchaeology | UE5, C++, Blueprint

May 2025 – Nov 2025

- Designed and implemented a modular quest system with editor tooling using C++ to guide user actions through reusable scripting logic and contextual prompts
- Developed an interactive, tool-based VR excavation system using C++, simulating surface digging and stratigraphic layer analysis
- Implemented shared sublevel instances in UE5 to maintain continuity across multiple uses of a shared environment, ensuring consistent gameplay and scene behavior, and simplifying the workflow for content teams

Multiplayer Action Game | UE5, C++, Blueprint, ASIO, C#, GraphQL, PostgreSQL

Nov. 2025 – Present

- Built a multiplayer combat framework using C++ and Epic's Gameplay Ability System, implementing client-side prediction and correction for movement abilities
- Developed a matchmaking utility service using C#, PostgreSQL database, and GraphQL API, integrating with Epic Online Services to let matchmaking use game state information without relying on UE Beacons
- Created a system for generating replicated procedural environments using Epic's new Procedural Content Generation framework

Card/Equipment Battler | UE5, C++, Blueprint

Jan 2024 – Jun 2024

- Implemented lightweight effect-based card and item combat system with card resource restrictions and item skills
- Created tooling to fetch and generate card assets and enemies from design spreadsheets
- Developed procedural link-based dungeon generation system including environment POIs and enemy encounters

Othello | Haskell, Python, OpenCV

May 2022

- Developed configurable Othello game with computer-controlled opponent using minimax algorithm and pruned game state tree using Haskell
- Implemented computer vision tool to determine Othello board state from images using Hough transform, utilizing Python and the OpenCV library

TECHNICAL SKILLS

Languages: C++, C, C#, SQL, Python, Haskell, Typescript, OpenGL, WebGL, Blueprint

Developer Tools: Unreal Engine, Perforce, Git, Visual Studio, Unity, Docker, Horde, Jenkins

Libraries: ASIO (Boost), Hot Chocolate(GraphQL), OpenCV, SlateIM, Numpy, Matplotlib